

“Online Advertisement System”

AN INTERNSHIP REPORT

Submitted by

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DIPLOMA ENGINEERING

in

INFORMATION TECHNOLOGY



College of Technology

Aditya Silver Oak Institute of Technology



Silver Oak University, Ahmedabad

April, 2025



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CERTIFICATE

This is to certify that the Internship report submitted along with the Internship entitled **Online Advertisement System** has been carried out by **Shaikh Kashifaanjum Iqbalahmed** under my guidance in partial fulfillment for the Diploma Engineering in IT, 6th Semester of Silver Oak University, Ahmedabad during the academic year 2025-26

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DECLARATION

We hereby declare that the Internship report submitted along with the Internship entitled **Online Advertisement System** submitted in partial fulfillment for the Diploma Engineering in Information Technology to Silver Oak University, Ahmedabad, is a bonafide record of original project work carried out by me at **Grownited Private Limited** under the supervision of **Rahul Kirpekar** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

ज्ञानं परमं भूषणम्

Name of the Student

Sign of Student

ACKNOWLEDGEMENT

The Internship opportunity I had with GROWNITED PRIVATE LIMITED was a great chance for learning and professional development. Therefore, I consider myself as a very lucky individual as I was provided with an opportunity to be a part of it. I am also grateful for having a chance to meet so many wonderful people and professionals who led me through this Internship period. I express my deepest thanks to Mr. Rahul Kirpekar, for giving necessary advices and guidance and arranged all facilities to make the things easier. I choose this moment to acknowledge his contribution gratefully. And last not the least I would like to express my deep sense of gratitude to all the facilities of Ahmedabad Institute of Technology whose helping hand was always there. I perceive as this opportunity as a big milestone in my career development. I will strive to use gained skills and knowledge in the best possible way, and I will continue to work on their improvement, in order to attain desired career objectives. Hope to continue cooperation with all of you in the future.

ABSTRACT

This report details the development and implementation of an Online Advertisement System, designed to streamline outdoor advertising through a structured and automated hoarding booking platform. The system provides a centralized solution for advertisers, administrators, and customers to manage advertisements efficiently while ensuring seamless booking, payment, and content management.

The system consists of several key modules, including User Management, which classifies users into customers and administrators, enabling role-based access and authentication. The Location Management Module maintains structured data on states, cities, and areas, ensuring accurate hoarding placement. The Hoarding Management Module tracks hoarding availability, dimensions, and pricing, allowing advertisers to select suitable locations based on their campaign needs. The Booking System facilitates hoarding reservations, enabling users to schedule advertisements for specific durations while calculating total costs. Additionally, the Payment Processing Module ensures secure transactions with multiple payment methods and real-time status tracking.

To enhance user experience and business efficiency, the system incorporates ad content storage, banner management, and real-time status updates of hoardings. Future enhancements include digital ad integration, SEO optimization, and advanced data analytics to provide better audience targeting and campaign performance insights. This system not only simplifies outdoor advertising but also modernizes the process, making it more accessible, efficient, and data-driven.

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CHAPTER: 1

OVERVIEW OF THE COMPANY

1.1 History

Grownited Private Limited is an open-source organization dedicated to fostering innovation and entrepreneurship. We believe that ideas are valuable, and our mission is to transform them into reality. By bridging the gap between visionary thinkers and skilled professionals, we cultivate startups and projects that shape a modern and futuristic India. As a start-up cultivation firm, we collaborate with idea cultivators, entrepreneurs, and innovators to bring groundbreaking concepts to life. Our goal is to become a hub of creativity, empowering young entrepreneurs and professionals to turn their visions into tangible solutions. At **Grownited Private Limited**, we recognize that experience is the key to success. That's why we offer internships in over 15 categories for individuals aged 14–26, providing them with the hands-on learning and mentorship needed to thrive in today's competitive landscape. Through our expertise in mobile app development, e-commerce solutions, website design, software development, and digital marketing, we deliver cutting-edge services that help businesses and individuals achieve their goals. With a passion for innovation and a commitment to excellence, **Grownited Private Limited** is not just a company—it's a movement towards a smarter, more connected, and entrepreneur-driven future.

1.2 Different work/ scope of Work:

1. Web Development
2. Mobile app development
3. E-commerce store development
4. Website Development
5. Backend Development
6. Software Development

1.3 Organization Chart

While specific details about Grownited's organizational structure are not publicly disclosed, the company emphasizes a collaborative approach, bringing together focused individuals who thrive on challenges to transform the business, creative, and technological realms.

1.4 Capacity of Plant

Grownited operates from its office located at 301-305, 3rd Floor, Surbhi Complex, Near Municipal Market, Chimanlal Girdharlal Road, Navrangpura, Ahmedabad, Gujarat - 380009. Specific information regarding the company's production capacity or physical plant facilities is not publicly available.

CHAPTER: 2 OVERVIEW OF DIFFERENT DEPARTMENT

Grownited Private Limited is structured into several key departments, each dedicated to delivering specialized services that collectively contribute to the company's mission of transforming innovative ideas into tangible solutions.

2.1 Departmental Functions and Activities

❖ Software Development Department:

- Functions:
 - Designing and developing custom software solutions tailored to client requirements.
 - Ensuring software quality through rigorous testing and maintenance protocols.
- Activities:
 - Requirement analysis and system design.
 - Coding and implementation using various programming languages.
 - Conducting unit and integration testing.
 - Deploying software and providing ongoing support.

❖ E-Commerce Development Department:

- Functions:
 - Creating e-commerce platforms for both B2B and B2C markets.
 - Integrating secure payment gateways and user-friendly interfaces.
- Activities:
 - Developing responsive and scalable online stores.
 - Implementing product catalog management systems.
 - Ensuring compliance with e-commerce regulations and standards.

❖ Mobile App Development Department:

- Functions:
 - Designing and developing mobile applications, primarily for the Android platform.
 - Ensuring apps are user-friendly, efficient, and meet client specifications.

- Activities:
 - Creating app prototypes and wireframes.
 - Coding using languages such as Java or Kotlin.
 - Testing applications for performance and security.
 - Deploying apps to the Google Play Store and providing maintenance updates.

❖ Website Design Department:

- Functions:
 - Crafting visually appealing and functional websites that effectively communicate client messages.
- Activities:
 - Developing site layouts and user interfaces.
 - Incorporating graphic design elements and multimedia content.
 - Ensuring cross-browser compatibility and mobile responsiveness.
 - Optimizing websites for search engines (SEO).

❖ Digital Marketing Department:

- Functions:
 - Promoting brands and products through various online channels to enhance visibility and engagement.
- Activities:
 - Developing and executing marketing strategies across platforms like social media, email, and search engines.
 - Analysing market trends and customer behaviour to inform campaigns.
 - Creating and curating content that aligns with brand messaging.
 - Monitoring and reporting on campaign performance metrics.

❖ Branding Department:

- Functions:
 - Establishing and promoting unique brand identities for clients to create a memorable market presence.
- Activities:
 - Designing logos, taglines, and brand guidelines.
 - Developing brand strategies that align with client visions and market positioning.
 - Conducting market research to understand brand perception.

Collaborating with other departments to ensure consistent brand messaging across all platforms.

2.2 Technical Specifications of Major Tools and Technologies Used

While specific equipment details are not publicly disclosed, Grownited Private Limited utilizes a range of industry-standard tools and technologies across its departments:

1. Software Development:

- Programming Languages: Python, Java, C++, JavaScript.
- Frameworks: Spring, Hibernate, React, Angular.
- Databases: MySQL, PostgreSQL, MongoDB.
- Version Control: Git, GitHub, Bitbucket.

2. E-Commerce Development:

- Platforms: Magento, Shopify, WooCommerce.
- Payment Gateways: PayPal, Stripe, Razorpay.

3. Mobile App Development:

- Development Environments: Android Studio.
- Languages: Java, Kotlin.
- Testing Tools: JUnit, Espresso.

4. Website Design:

- Design Tools: Adobe Photoshop, Illustrator, Figma.
- Web Technologies: HTML5, CSS3, JavaScript, Bootstrap.

5. Digital Marketing:

- SEO Tools: Google Analytics, SEMrush, Ahrefs.

- Social Media Platforms: Facebook, Instagram, LinkedIn, Twitter.
- Email Marketing: Mailchimp, SendGrid.

6. Branding:

- Design Software: Adobe Creative Suite.
- Market Research Tools: SurveyMonkey, Brandwatch.

2.3 Schematic Layout of Service Delivery Process

As a service-oriented company, Grownited follows a structured process to deliver its solutions:

1. Client Consultation:

- Understanding client needs, objectives, and challenges.
- Gathering initial requirements and expectations.

2. Proposal Development:

- Crafting a tailored solution plan, including scope, timelines, and cost estimates.
- Presenting the proposal to the client for approval.

2.4 Detailed Explanation of Each Stage of Production

1. Requirement Gathering

- The client shares their project requirements.
- Business analysts assess feasibility and propose solutions.

2. Planning & Design

- Project managers define scope, budget, and timeline.
- UI/UX designers create wireframes and prototypes.

3. Development Phase

- Software engineers build the required application or service.
- Web and mobile developers code interfaces and backend logic.

4. Testing & Quality Assurance

- Unit, integration, and performance tests are conducted.
- Security testing ensures protection against cyber threats.

5. Deployment & Launch

- The software/app/website is deployed on production servers.
- Final optimizations and bug fixes are made before public access.

6. Marketing & Branding

- SEO strategies, social media ads, and promotional campaigns are executed.
- Branding elements (logos, themes) are incorporated into the digital assets.

7. Post-Launch Support & Maintenance

- Regular updates and security patches are applied.
- Client feedback is collected to improve future versions.

CHAPTER: 3 INTRODUCTION

3.1 Project Summary

1. Introduction to Online Advertisement System Project

The Online Advertisement System is designed to modernize and streamline the process of outdoor advertising by providing a digital platform for booking and managing hoardings. This project focuses on automating the advertisement placement process, making it easier for businesses to promote their products while ensuring efficient utilization of hoarding spaces.

The system allows advertisers to book hoardings in various locations, manage their advertisements, and track campaign performance. Administrators oversee hoarding availability, bookings, and payments, ensuring smooth operations. The project integrates user management, hoarding tracking, booking system, and payment processing, creating a comprehensive and efficient advertising solution.

This project serves as a practical application of database management, web technologies, and digital marketing concepts, making it an ideal internship or project for students and professionals interested in advertising technology and business automation.

2. Project / Internship Management

Managing the Online Advertisement System project involves a structured approach to ensure efficient development, implementation, and maintenance. The key aspects of project management include:

Project Planning & Objective Setting: Defining the system's scope, features, and functionalities, including user roles (admin, customer), hoarding booking, and payment integration.

Task Allocation & Development Process: Assigning responsibilities for database design, front-end and back-end development, and testing.

Resource Management: Utilizing programming languages, frameworks, and database management tools to ensure smooth system functionality.

Monitoring & Evaluation: Regular testing to track system performance, identify bugs, and improve user experience.

Reporting & Documentation: Maintaining detailed records of system design, user workflows, challenges, and implemented solutions for future improvements.

3.2 Purpose

The purpose of the Online Advertisement System is to digitize and automate the process of outdoor advertising by providing a structured and efficient platform for managing hoardings. Traditional hoarding advertisement booking is often time-consuming, manual, and lacks transparency in terms of availability, pricing, and campaign tracking. This system eliminates these inefficiencies by offering a centralized, user-friendly, and automated solution for advertisers and hoarding providers.

The key objectives of this project are:

Simplifying Advertisement Booking – Enabling advertisers to easily browse available hoardings, select locations, schedule ad campaigns, and book spaces in real-time.

Efficient Hoarding Management – Helping administrators manage hoarding availability, pricing, and bookings to ensure optimal utilization of advertising spaces.

Automating Payment Processing – Providing a secure and transparent payment system with multiple payment methods and real-time transaction tracking.

Enhancing User Experience – Offering an intuitive interface for both advertisers and administrators, reducing the need for manual intervention.

Ensuring Data Accuracy – Maintaining a structured database of states, cities, areas, hoardings, and bookings, preventing errors and duplications.

Improving Ad Campaign Effectiveness – Allowing advertisers to upload ad content, manage banners, and track their campaign performance efficiently.

Future Scalability – Enabling future enhancements such as digital ad integration, SEO optimization, and data analytics for better audience targeting and campaign insights.

By achieving these goals, the Online Advertisement System not only optimizes the outdoor advertising industry but also provides a modern, cost-effective, and scalable solution that benefits businesses, advertisers, and hoarding providers alike.

3.3 Objective

The Online Advertisement System aims to revolutionize the traditional outdoor advertising process by automating hoarding bookings, streamlining payments, and enhancing ad management. The system is designed to provide an efficient, user-friendly, and scalable solution for advertisers, administrators, and hoarding providers.

The key objectives of this project include:

Develop a Centralized Platform – Create a web-based system where advertisers can browse, book, and manage hoardings with ease.

Automate the Hoarding Booking Process – Enable users to check availability, reserve hoardings, and schedule advertisements in real-time.

Enhance User Management – Offer role-based authentication for customers and administrators, ensuring proper access control and security.

Efficient Location-Based Advertisement Placement – Integrate a structured database of states, cities, and areas to help advertisers select the best hoarding locations.

Enable Ad Content Management – Allow users to upload, update, and manage advertisement banners for hoardings.

Optimize Hoarding Utilization – Help administrators manage hoarding availability, pricing, and performance tracking to maximize revenue.

Improve Business Efficiency – Reduce manual effort in booking, payment handling, and campaign monitoring, making the process more cost-effective and time-efficient.

Ensure Scalability and Future Expansion – Design the system to support future enhancements such as digital ads, SEO optimization, and advanced analytics for better audience targeting.

By fulfilling these objectives, the Online Advertisement System ensures a seamless, automated, and data-driven approach to outdoor advertising, benefiting both advertisers and hoarding providers while modernizing the industry.

3.4 Scope

The Online Advertisement System is designed to streamline the process of booking and managing outdoor hoardings through an automated digital platform. It provides a structured workflow for advertisers to book ad spaces, upload content, and track campaigns, while administrators manage hoarding availability, bookings, and payments. However, like any system, it has defined capabilities and limitations.

❖ Scope of the System (What It Can Do)

1. User Management:
 - Supports two roles: Admin and Customer.
 - Allows registration, login, and profile management for users.
 - Stores user details such as name, contact, and role.
2. Location-Based Advertisement Management:
 - Stores state, city, and area details where hoardings are available.
 - Users can search and filter hoardings based on location.
3. Hoarding (Billboard) Management:
 - Allows businesses to list and manage hoardings in different cities.
 - Each hoarding has details such as size, location, price, and availability status.
 - Users can view sample templates of how the hoarding will look.
4. Booking System:
 - Customers can book available hoardings for advertising.
 - The system tracks start and end times for each booking.
 - Customers can upload advertisement banners.
5. Admin Features:
 - Admins can manage hoardings, bookings, users, and payments.
 - View reports on total users, monthly bookings, and total banners displayed.

- Monitor advertisement trends and customer preferences.

❖ **Limitations (What It Can't Do)**

1. Does not handle real-time dynamic pricing – Currently, the system follows a fixed rate per hoarding instead of dynamic pricing based on demand.
2. No real-time hoarding availability updates – Hoarding availability must be updated manually by the admin.
3. Limited to offline hoardings – The system does not support digital or online ad placements like Google Ads or social media advertising.
4. No automated content moderation – The system does not check or approve advertisement content automatically; this must be done manually by the admin.
5. Limited analytics dashboard – While basic reports are available, advanced data visualization and ad performance tracking are not included.

3.5 Technology and Literature Review

The Online Advertisement System is developed using Java-based technologies to ensure a scalable, secure, and efficient platform for managing outdoor advertising hoardings. The system follows a three-tier architecture, separating the frontend, backend, and database for better maintainability and performance.

❖ Frontend Technologies:

- JSP (JavaServer Pages) – Used for creating dynamic web pages with embedded Java code.
- HTML, CSS, JavaScript – Ensures a responsive and interactive user interface.
- Bootstrap – Provides a modern and user-friendly UI design.

❖ Backend Technologies

- Java Spring Boot – A powerful backend framework that simplifies development, provides RESTful APIs, and ensures high performance.
- Spring MVC – Manages application requests efficiently using the Model-View-Controller (MVC) architecture.
- Spring Security – Ensures secure authentication and authorization for users.

❖ Database Management System (DBMS)

- MySQL – A relational database used for storing user details, hoarding information, bookings, and payment records.
- JPA (Java Persistence API) / Hibernate – Manages database interactions and ensures efficient ORM (Object-Relational Mapping).

❖ Server & Deployment

- Apache Tomcat – A lightweight and robust server used for deploying and running JSP and Spring Boot applications.
- Cloud Hosting (AWS, Google Cloud) – Can be used for scalability and remote access.

❖ **Authentication & Security**

- Spring Security with JWT (JSON Web Token) – Provides secure user authentication.
- SSL Encryption – Protects sensitive user data (passwords, transactions).
- Payment Gateway Integration
- Razorpay / PayPal / Stripe – Allows users to make secure online payments for booking hoardings.

Literature Review

The Online Advertisement System is inspired by existing advertisement management platforms and business automation models. Several research papers, case studies, and commercial applications have contributed to the design of this system.

1. Importance of Automation in Outdoor Advertising

- Traditional hoarding booking systems are manual and inefficient, leading to delays, mismanagement, and revenue loss.
- Automated solutions like Google Ads have revolutionized digital marketing, but outdoor advertising still requires structured database-driven automation.
- This project bridges the gap by providing an automated online platform for hoarding booking and management.

2. Java Spring Boot for Scalable Web Applications

- Research on Spring Boot applications shows that it reduces development complexity, making it ideal for large-scale applications.
- Using Spring MVC and Hibernate, the system ensures faster development, database efficiency, and seamless API communication.

3. Database-Driven Ad Management Systems

- Studies highlight that structured databases improve ad placement efficiency.
- The system leverages MySQL to manage states, cities, hoardings, bookings, and payments for optimized advertisement targeting.

4. Security & Payment Processing in Online Booking Systems

- Research on online booking fraud prevention emphasizes the importance of secure authentication and encrypted transactions.
- The system uses Spring Security and JWT for user verification, and SSL encryption for secure payments and transactions.

3.6 Project / Internship Planning

3.6.1 Project / Internship Development Approach and Justification

The Online Advertisement System follows the Agile Software Development Methodology, specifically the Iterative Model, to ensure flexibility, continuous improvement, and faster delivery. This approach allows the development team to deliver working prototypes in cycles, gather feedback, and refine the system before final deployment.

1. Development Approach: Agile Methodology (Iterative Model)

The Agile approach divides the development process into multiple iterations or "sprints," where each sprint delivers a functional module of the system.

❖ Phases of Agile Development for this Project:

Phase Tasks Justification

1. Requirement Gathering & Analysis

- Identify system scope
- user roles (Admin, Customer)
- core functionalities.
- Ensures a clear understanding of project goals before development begins.

2. System Design & Database Modeling

- Define architecture
- database schema (MySQL)

3. API endpoints

- Helps create a well-structured system for efficient data management and performance.

4. Backend Development (Spring Boot)

- Develop RESTful APIs for authentication
- booking, payments
- hoarding management.
- Ensures a scalable and secure backend using Spring Boot and Hibernate.

5. Frontend Development (JSP, Bootstrap)

- Build interactive UI for advertisers and admin to book hoardings, manage ads, and make payments.
- Provides a user-friendly experience for customers and administrators.

6. Integration & Testing

- Connect frontend with backend, run unit tests and integration tests.
- Detects bugs and performance issues early in development.

7. Deployment & User Feedback

- Deploy on Apache Tomcat
- collect user feedback
- implement enhancements.
- Ensures a stable system before final release.

2. Justification for Using Agile Development

1. Changes in requirements, design, or features can be incorporated in each sprint.

- Suitable for projects where requirements may evolve over time.

2. Faster Delivery & Continuous Improvement

- Working prototypes are delivered after each iteration, ensuring progress is visible.
- System improves continuously with each sprint, leading to a high-quality final product.

3. Early Bug Detection & Risk Management

- Regular testing in each sprint helps detect and fix issues early.
- Reduces risks related to performance, security, and functionality.

4. User-Centric Development

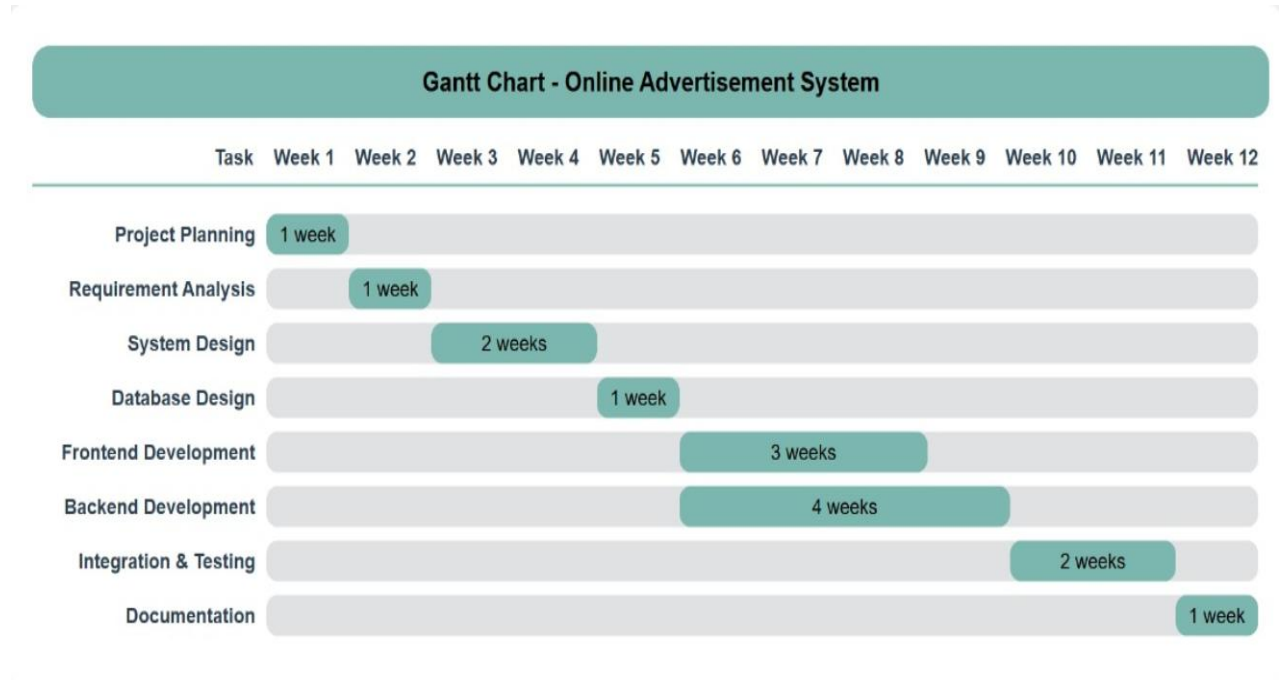
- Developers receive early user feedback and refine features accordingly.
- Ensures the system is aligned with real-world needs.

5. Better Collaboration & Transparency

- Agile promotes team collaboration, ensuring all stakeholders (developers, clients) stay involved.
- Progress tracking through daily stand-ups and sprint reviews.

3.6.2 Project / Internship Effort and Time

Project Time:



3.6.3 Roles and Responsibilities

The Online Advertisement System involves different stakeholders, each with specific roles and responsibilities to ensure the smooth functioning of the platform. The key roles include Admins, Customers (Advertisers), and System Developers.

1. Admin

Responsibilities:

- Manage user accounts (approve, activate, or deactivate users).
- Oversee hoarding availability, including adding, updating, or removing hoardings.
- Monitor bookings and payments, ensuring transactions are completed.
- Approve advertisement content before display.
- Generate reports on total users, bookings, payments, and system performance.
- Handle customer queries and support requests.

2. Customer (Advertiser)

Responsibilities:

- Register on the platform and manage their user profile.
- Search and book available hoardings based on location, size, and cost.
- Upload advertisement content and banners for display.
- Make secure payments using available payment methods.
- Track booking history and advertisement performance.
- Request modifications or cancellations of bookings (if applicable).

3. System Developer (Development Team)

Responsibilities:

- Design and implement the backend using Java Spring Boot.
- Develop the frontend using JSP and ensure a user-friendly interface.
- Manage the database (MySQL) to store user, booking, and payment data,
- Integrate secure payment gateways for smooth transactions.
- Deploy and maintain the server using Tomcat.
- Fix bugs, update security features, and optimize system performance.

4. Database Administrator (DBA)

Responsibilities:

- Design and manage the database schema to store advertisement, booking, and payment details efficiently.
- Ensure data security, backup, and recovery mechanisms.
- Optimize database queries and indexing for faster performance.
- Maintain user authentication and access control for data protection.

5. Tester (Quality Assurance)

Responsibilities:

- Perform functional, performance, and security testing.
- Identify and report bugs, UI issues, and payment failures.
- Conduct user acceptance testing (UAT) before deployment.
- Ensure system reliability and error-free performance.

3.6.4 Group Dependencies

In the Online Advertisement System, various teams and stakeholders depend on each other to ensure smooth operation, development, and maintenance of the platform. These dependencies define how different roles interact and rely on one another.

1. Admin ↔ Customer Dependency

- Admins manage the platform, while customers use it.
- Customers depend on admins to approve bookings, verify payments, and resolve issues.
- Admins rely on customers to provide accurate ad content, complete payments, and follow guidelines.

2. Customer ↔ Hoarding Availability Dependency

- Customers rely on the availability of hoardings to book advertisements.
- If hoardings are fully booked or unavailable, customers cannot place ads.
- The system must regularly update hoarding availability to reflect real-time status.

3. Admin ↔ System Developer Dependency

- Admins need the system to function correctly, and developers maintain it.
- If a technical issue occurs (e.g., payment failure, booking error), admins depend on developers to fix it.
- Developers require admin feedback to improve features, fix bugs, and update the system.

4. Developer ↔ Database Administrator (DBA) Dependency

- Developers rely on the DBA to manage the database efficiently.
- Developers need structured and optimized data storage for smooth system performance.
- The DBA ensures data security, backup, and integrity, which developers rely on for seamless functionality.

5. Developer ↔ Tester (Quality Assurance) Dependency

- Developers build the system, while testers ensure it works correctly.
- Testers identify bugs, performance issues, and UI errors.
- Developers rely on testers' reports to debug and optimize the system before deployment.

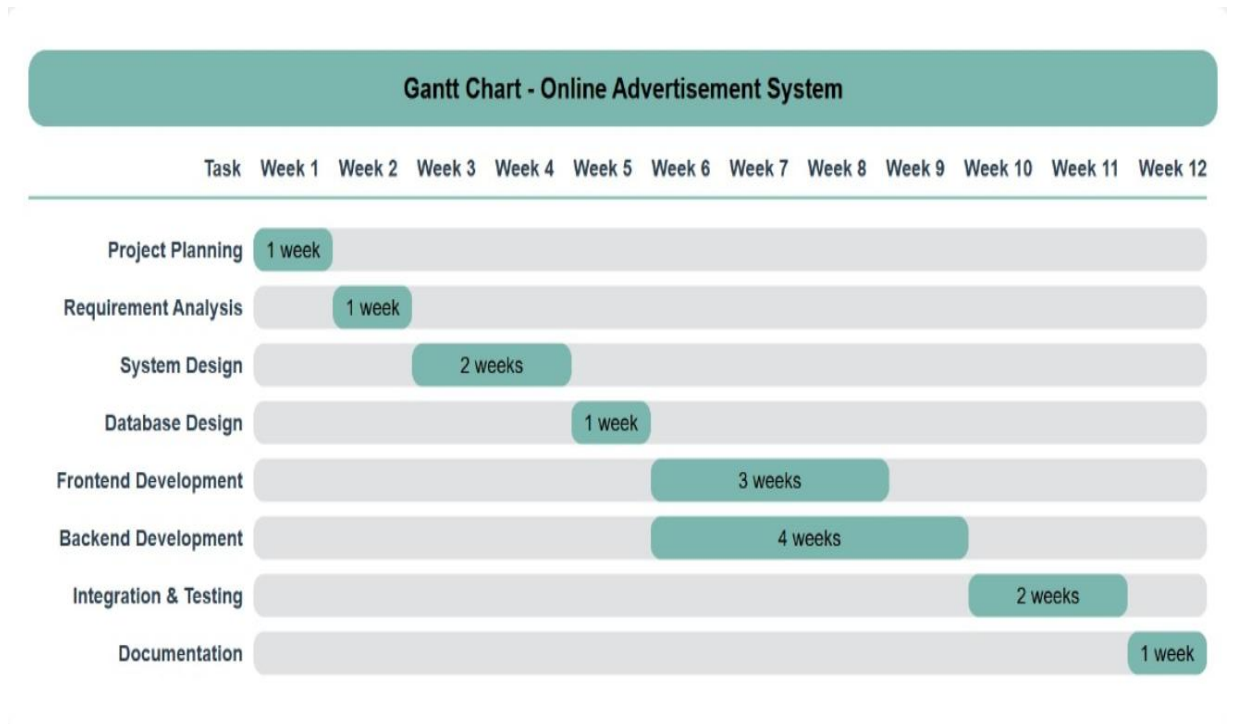
6. Payment Gateway ↔ Customer Dependency

- Customers depend on the payment gateway to complete transactions.
- If a payment gateway fails, customers cannot confirm bookings.
- The system must ensure secure and smooth transactions for hassle-free payments.

7. Project Manager ↔ Entire Team Dependency

- The Project Manager coordinates tasks and ensures smooth execution.
- The PM depends on developers, testers, DBAs, and admins to complete their tasks on time.
- Team members rely on the PM for task allocation, resource management, and decision-making.

3.7 Project / Internship Scheduling (Gantt Chart)



CHAPTER:4 SYSTEM ANALYSIS

Introduction

The Online Advertisement System is designed to facilitate the efficient booking and management of hoardings in various cities. This system enables users, including customers and administrators, to manage advertising spaces seamlessly. The platform incorporates user management, booking, payment processing, and future scope for digital and SEO advancements.

❖ System Components

1. User Management

- The system maintains records of different types of users, including customers and administrators.
- User Table: Stores personal details such as first name, last name, email, contact number, and role (admin/customer).
- Admin: Manages the system, hoardings, and bookings.
- Customer: Can browse and book hoardings.

2. Location Management

- The system organizes hoardings by state, city, and area to facilitate location-based searches.
- State Table: Contains unique identifiers and names for states.
- City Table: Links to states and stores city-specific data.
- Area Table: Maps areas within cities for more granular hoarding placements.

3. Hoarding Management

- The system tracks available and booked hoardings with details such as dimensions, location, availability, pricing, and associated templates.
- Hoarding Table: Stores attributes like hoarding ID, address, dimensions, availability status, hourly rate, and a URL template for the advertisement.
- Location Mapping: Each hoarding is linked to a state, city, and area.

4. Booking Management

- Customers can book available hoardings for advertising purposes.
- Booking Table: Records details such as booking ID, client ID, hoarding ID, start and end times, total cost, payment status, and content details.
- Ad Content Management: Allows customers to upload advertisement banners.

5. Payment Management

- The system ensures secure payment transactions for bookings.
- Payment Table: Includes payment ID, client ID, booking ID, amount, payment method, and transaction reference.
- Payment Status Tracking: Records whether payments are completed or pending.

6. Features

- User Authentication & Role-Based Access - Customers and admins have distinct permissions.
- Search & Filter Hoardings - Users can filter based on location, availability, and pricing.
- Real-Time Availability - Ensures that users book only available hoardings.
- Automated Pricing & Cost Calculation - Calculates total booking costs based on duration.

- Secure Payment Gateway - Supports multiple payment methods like credit cards and cash.

7. Future Scope

- Digital Hoardings - Integration of LED and digital advertising spaces.
- SEO Optimization - Enhancing discoverability of hoarding spaces through search engine optimization.
- Automated Campaign Management - AI-driven ad placement recommendations.
- Analytics & Reporting - Insights on advertisement performance and booking trends.

4.1 Study of Current System

The current advertisement booking system is largely manual and fragmented, leading to inefficiencies in hoarding management. Several issues hinder the smooth functioning of traditional advertisement booking methods:

❖ Challenges in the Current System

1. Lack of Centralized Management

- Hoardings are managed independently by different agencies, making it difficult for advertisers to find and book available spaces.
- There is no unified platform to manage hoarding inventory across multiple cities.

2. Limited Availability Tracking

- Advertisers do not have real-time access to hoarding availability, leading to issues like overbooking or missed opportunities.
- Manual coordination between agencies and advertisers often results in delays and inefficiencies.

3. Manual Payment Processing

- Most transactions are conducted offline, increasing the risk of financial discrepancies and fraud.
- Delayed payment confirmations can hinder smooth booking and ad deployment.

4. Inefficient Advertising Space Utilization

- Without proper data insights and analytics, advertisers struggle to choose the best locations for their campaigns.

- The lack of a structured approach to space utilization leads to revenue loss for hoarding owners.

5. Poor Customer Experience

- Advertisers face difficulty in comparing hoarding options, negotiating prices, and ensuring timely booking and ad placement.
- The absence of an integrated digital platform means advertisers rely on lengthy manual processes to secure advertising spots.

6. Need for an Improved System

- Given the challenges above, there is a clear need for a centralized and automated system that:
- Provides real-time availability tracking of hoardings.
- Enables secure and efficient online payment processing.
- Offers data-driven insights for advertisers to optimize their campaigns.
- Streamlines communication between advertisers and hoarding providers.
- Enhances customer experience through a seamless booking process.

The Online Advertisement System aims to address these inefficiencies by providing an integrated platform that simplifies hoarding management, ensures transparency, and enhances operational efficiency.

4.2 Problems and Weaknesses of the Current System

The current system suffers from multiple drawbacks that negatively impact the efficiency of advertisement bookings. Key weaknesses include:

1. Time-Consuming Manual Process

- Booking hoardings requires direct communication with agencies, leading to long processing times.
- Approval and confirmation of bookings can take days or weeks.

2. Lack of Data Insights

- Advertisers do not have access to analytical data that could help optimize campaign placement.
- Agencies lack tools to monitor the effectiveness of hoardings and maximize revenue.

3. High Risk of Errors

- Manual record-keeping increases the chances of booking conflicts and incorrect pricing.
- The absence of digital tracking makes it difficult to resolve disputes efficiently.

4. Limited Payment Options

- Advertisers are often restricted to cash or bank transfers, making transactions inconvenient.
- There is no automated invoicing or payment tracking system.

These weaknesses demonstrate the necessity of a robust, automated, and centralized system to streamline operations and improve user experience.

4.3 Requirements of the New System

To address the problems in the current system, the new Online Advertisement System must incorporate the following features:

❖ Functional Requirements

1. User Authentication and Role-Based Access

- Secure login for customers and administrators.
- Role-based permissions to ensure controlled access.

2. Hoarding Management System

- A centralized database for all hoarding locations, dimensions, and pricing.
- Real-time tracking of hoarding availability.

3. Automated Booking System

- Instant booking confirmation and digital contracts.
- Scheduling features to prevent booking conflicts.

4. Payment Integration

- Support for online payments via credit/debit cards, net banking, and digital wallets.
- Automated invoicing and transaction tracking.

5. Data Analytics and Reporting

- Insights into hoarding performance based on views, location, and past bookings.
- Advertisers can analyze their campaign reach and effectiveness.

6. Customer Support and Dispute Resolution

- A built-in support system for addressing complaints and queries.
- Clear policies for refunds and booking modifications.

❖ Non-Functional Requirements

1. Scalability

- The system should handle multiple users and transactions simultaneously without performance issues.

2. Security

- Strong encryption for user data and payment details.
- Role-based access control to protect sensitive information.

3. User-Friendly Interface

- Intuitive navigation for advertisers and administrators.
- Mobile compatibility for on-the-go access.

By implementing these requirements, the new system will provide a seamless and efficient experience for all users.

4.4 System Feasibility

Before implementing the Online Advertisement System, it is essential to evaluate its feasibility in terms of various parameters:

1. Technical Feasibility

- The system can be developed using modern web technologies such as cloud databases, secure APIs, and responsive design frameworks.
- Integration with payment gateways and data analytics tools is feasible.

2. Economic Feasibility

- Initial development costs include software development, server hosting, and API integrations.
- Revenue generation through commission-based bookings and premium ad placements ensures long-term sustainability.
- Automation reduces administrative costs by minimizing manual work.

3. Operational Feasibility

- Agencies and advertisers can quickly adopt the system with minimal training.
- A user-friendly interface ensures that even non-technical users can navigate the platform effectively.
- Real-time updates improve efficiency and transparency in the booking process.

4. Legal and Regulatory Feasibility

- The system will comply with digital transaction laws and data protection regulations.
- Terms of service will outline advertiser responsibilities and agency policies.

- Legal contracts will be auto-generated for each booking to ensure clarity and enforceability.

By assessing these feasibility aspects, it is clear that the proposed system can be successfully implemented to transform the advertisement booking industry.

4.4.1 Does the System Contribute to the Overall Objectives of the Organization?

Yes, the proposed Online Advertisement System aligns with the overall objectives of the organization in several ways:

1. Enhancing Efficiency and Automation

- The system automates booking, payment processing, and availability tracking, reducing manual workload and minimizing errors.
- Real-time data access improves operational efficiency and ensures quick decision-making.

2. Revenue Growth and Cost Optimization

- By optimizing space utilization and reducing booking conflicts, the system maximizes revenue for hoarding providers.
- Automation minimizes administrative costs, increasing overall profitability.

3. Improving Customer Experience

- A seamless booking and payment process enhances user satisfaction.
- Transparent pricing and availability tracking provide a hassle-free experience for advertisers.

4. Competitive Advantage

- The system provides advanced analytics, helping advertisers make data-driven decisions.
- Digital integration and automation give the organization a technological edge over competitors.

5. Compliance and Security

- Ensuring secure transactions and adherence to data protection regulations builds trust among users.
- Digital contracts and automated invoicing improve financial transparency.

By addressing these key areas, the system directly contributes to the organization's goals of efficiency, profitability, customer satisfaction, and competitive growth.

4.4.2 Can the System Be Implemented Using the Current Technology and Within the Given Cost and Schedule Constraints?

The feasibility of implementing the Online Advertisement System depends on technological availability, budget constraints, and time limitations. Analyzing these factors:

1. Technological Feasibility

- The system can be developed using modern web technologies such as cloud databases, secure APIs, and mobile-friendly platforms.
- Integration with payment gateways, real-time availability tracking, and automated reporting is achievable with existing technologies.
- The use of scalable cloud infrastructure ensures the system can handle increasing traffic and user demand efficiently.

2. Economic Feasibility

- The estimated cost of development includes software design, development, hosting, and maintenance.
- Revenue generation through booking commissions and premium ad placements ensures financial sustainability.
- The initial investment in development is justified by long-term cost savings and increased revenue streams.

3. Time and Schedule Feasibility

- The system can be implemented within the planned schedule by following an agile development approach.
- Modular development ensures core functionalities like booking, payments, and analytics can be launched first, with additional features added incrementally.
- Adequate resource allocation and risk management will help avoid delays in implementation.

4. Operational Feasibility

- Minimal training is required for advertisers and hoarding providers to use the system effectively.
- A user-friendly interface ensures smooth adoption by all stakeholders.
- Regular updates and customer support will enhance operational efficiency post-deployment.

Considering these factors, the Online Advertisement System can be successfully implemented using current technology and within the given cost and schedule constraints, making it a viable solution for the organization.

4.4.3 Can the System Be Integrated with Other Systems Which Are Already in Place?

The integration of the Online Advertisement System with existing systems is essential for ensuring seamless data flow and operational efficiency. The feasibility of integration is analyzed based on the following factors:

1. Compatibility with Existing Systems

- The system is designed with API support to ensure easy integration with existing enterprise software such as CRM, billing systems, and marketing tools.
- Database structures will be aligned with existing platforms to facilitate smooth data exchange.

2. Data Synchronization and Interoperability

- Real-time synchronization of bookings, payments, and advertisement content with other business management tools enhances efficiency.
- Secure data transfer protocols will be implemented to prevent inconsistencies and data duplication.

3. Integration with Payment Gateways

- The system can be linked with widely used payment platforms to ensure smooth financial transactions.
- Secure and standardized payment APIs will be used to maintain compliance with industry regulations.

4. Scalability and Future Integration Potential

- The modular architecture allows for future enhancements and easy integration with upcoming technologies.
- AI-driven analytics and digital marketing automation tools can be added without disrupting core functionality.

5. Minimal Disruptions During Integration

- A phased approach to system integration ensures minimal downtime and operational disruptions.
- User training and support will be provided to facilitate a smooth transition.

With these integration capabilities, the Online Advertisement System can be seamlessly incorporated into the organization's existing technological ecosystem, enhancing overall functionality and efficiency.

4.5 Activity / Process in New System / Proposed System

The proposed Online Advertisement System follows a structured workflow to ensure seamless operations. The key activities and processes involved are as follows:

1. User Registration and Authentication

- Advertisers and hoarding providers register on the platform and create their profiles.
- Secure authentication mechanisms ensure data protection and access control.

2. Hoarding Management

- Providers list available hoardings with details such as location, size, and pricing.
- Real-time availability updates prevent double bookings and optimize resource utilization.

3. Advertisement Booking Process

- Advertisers search for available hoardings and place booking requests.
- The system calculates total costs based on booking duration and hoarding rates.
- Automatic reservation confirmation ensures quick processing.

4. Payment Processing

- Secure payment gateway integration allows for multiple payment options.
- Payment status tracking (e.g., paid, pending) ensures financial transparency.
- Automated invoicing and receipts enhance record-keeping.

5. Advertisement Content Management

- Advertisers upload banners or digital content for display.
- The system validates content compliance with advertising guidelines.
- Hoarding providers approve or reject content before final display.

6. Reporting and Analytics

- The system generates real-time reports on ad performance, booking trends, and revenue analysis.
- AI-powered insights help advertisers optimize their campaigns.

7. Customer Support and Feedback

- Integrated support tickets and chatbots assist users with queries and technical issues.
- Feedback collection and review mechanisms improve service quality.

8. Future Enhancements

- The system is designed for scalability, allowing for the integration of digital hoarding solutions and AI-driven ad targeting.
- Advanced features such as programmatic advertising and automated bidding can be incorporated over time.

By implementing these streamlined processes, the new system enhances efficiency, ensures smooth operations, and improves user experience, making it a comprehensive solution for online advertisement management.

4.6 Features of the New System / Proposed System

The proposed system is designed to manage users, locations, hoardings, bookings, and payments efficiently. It includes robust data storage and retrieval functionalities, ensuring smooth operation for both administrators and customers.

1. User Management

- The system stores user data, including role-based access for administrators and customers. The key attributes are:
- User_Id: Unique identifier for each user.
- FirstName & LastName: Personal details of the user.
- Gender: User's gender.
- Email & Password: Authentication credentials.
- ContactNum: Contact details.
- CreatedAt: Account creation date.
- Active: Indicates whether the user is active.
- Role: Specifies whether the user is an admin or customer.

2. Location Management

- State
- Stores user state information:
- State_Id: Primary key.
- State_Name: Name of the state.
- City
- Stores user city details:
- City_Id: Primary key.
- City_Name: Name of the city.
- State_Id: Foreign key linking to the state.
- Area

- Stores area information:
- Area_Id: Primary key.
- Area_Name: Name of the area.
- State_Id & City_Id: Foreign keys linking to the respective state and city.

3. Hoarding Management

- The system handles hoardings efficiently, providing details about location and availability.
- Hoarding_Id: Primary key.
- Address: Location where the hoarding is placed.
- HoardingDimension: Specifies hoarding size (e.g., 10x10 feet).
- Availability_Status: Indicates availability (booked or available).
- HourlyRate: Cost per hour of usage.
- Hoarding_Url: Stores hoarding templates.
- State_Id, City_Id, Area_Id: Foreign keys to identify the exact location.
- HoardingTitle: Name of the hoarding.

4. Booking Management

- The system facilitates booking of hoardings, ensuring accurate tracking.
- Booking_Id: Primary key.
- Client_Id: Foreign key linking to the user.
- Hoarding_Id: Foreign key linking to the hoarding.
- Start_Time & End_Time: Booking duration.
- Total_Cost: Computed cost based on duration.
- Payment_Status: Indicates if the payment is completed or pending.
- AddContent: Stores basic ad content.
- AddBannerUrl: Stores the advertisement banner.

5. Payment Management

- Ensures seamless transactions and record-keeping.
- Payment_Id: Primary key.
- Client_Id: Foreign key linking to the user.
- Booking_Id: Foreign key linking to the booking.
- Amount: Payment amount.
- Payment_Method: Specifies method (credit card, cash, etc.).
- Payment_Status: Indicates payment completion.
- TransactionRef: Stores transaction reference.

6. Future Scope

- Digital Integration
- Implementation of digital platforms for hoarding advertisements.
- Online booking and management system enhancement.
- Search Engine Optimization (SEO)
- Optimizing content to improve visibility on search engines.
- Enhancing marketing strategies to attract more users.

This system provides a well-structured and efficient way to manage hoardings, ensuring ease of use and accessibility for users while streamlining the booking and payment processes.

4.7 List of Main Components of the New System

The proposed system consists of multiple modules that work together to provide seamless hoarding management, booking, and payment processing. The main modules include:

1. User Management Module

- Handles user registration, authentication, and authorization.
- Supports role-based access (Admin/Customer).
- Manages user profile updates and account activity status.

2. Location Management Module

- Stores and manages hierarchical location data (State, City, Area).
- Ensures hoardings are categorized accurately by geographical regions.

3. Hoarding Management Module

- Admins can add, update, and remove hoardings from the system.
- Stores hoarding attributes such as size, location, availability, and pricing.
- Provides real-time availability status updates.

4. Booking Management Module

- Allows users to check availability and book hoardings.
- Calculates total cost based on hourly or daily rates.
- Manages booking history and transaction details.

5. Payment Processing Module

- Supports multiple payment methods (Credit Card, UPI, Cash, etc.).
- Tracks payment status (Completed/Pending).
- Securely processes transactions with encryption.

6. Advertisement Content Management Module

- Enables users to upload and manage ad content for booked hoardings.
- Stores ad banners and display templates.
- Supports multiple media formats for ad placement.

7. Reporting and Analytics Module

- Generates reports on user activity, hoarding bookings, and revenue.
- Provides insights for admins to optimize hoarding placements and pricing.
- Supports graphical dashboards for data visualization.

❖ Components

1. Frontend Component

- Developed using JavaServer Pages (JSP) for dynamic web pages.
- Implements user-friendly interfaces for customers and administrators.
- Uses JavaScript and CSS for enhanced responsiveness.

2. Backend Component

- Built using Java Spring Boot for handling business logic.
- Implements RESTful APIs to communicate with frontend and database.
- Manages authentication, authorization, and role-based access.

3. Database Component

- Uses MySQL for structured data storage
- Ensures data integrity with relational mapping.
- Supports indexing for optimized query performance.

4. Server Component

- Hosted on Apache Tomcat Server for secure and efficient deployment.
- Manages concurrent user access with load balancing.
- Ensures high availability and reliability.

❖ Processes

1. User Authentication & Role-Based Access

- Uses Spring Security for secure login and role management.
- Implements password hashing with SHA-256 encryption.

2. Hoarding Availability & Booking Process

- Users search for hoardings based on location and availability.
- The system checks real-time booking status and calculates pricing.
- Users confirm bookings and proceed with payments.

3. Payment Processing Workflow

- Users select preferred payment methods (Credit Card, UPI, Cash).
- System securely processes transactions and updates booking status.
- Admins verify and manage completed transactions.

4. Advertisement Content Upload & Approval

- Customers upload advertisement banners for booked hoardings.
- Admins review and approve content before final placement.

5. Analytics & Reporting

- System generates insights on revenue, hoarding utilization, and user activity.
- Admins use reports for business decisions and optimizations.

❖ Techniques

1. Database Optimization Techniques

- Uses indexing for faster query execution.
- Implements foreign keys to ensure relational consistency.
- Enables data backup and recovery mechanisms.

2. Security Techniques

- Implements SSL encryption for secure transactions.
- Uses JWT (JSON Web Token) for user authentication.
- Prevents SQL injection and XSS attacks through data validation.

3. Performance Optimization Techniques

- Uses caching for frequently accessed data.
- Implements lazy loading for optimized resource management.
- Deploys load balancing for high-performance web service.

This structured approach ensures an efficient, secure, and scalable hoarding management system.

4.8 Selection of Hardware / Software / Algorithms / Methodology / Techniques / Approaches and Justification

1. Hardware Requirements

- Server: Dedicated or cloud-based Apache Tomcat server.
- Storage Devices: SSDs for fast data retrieval and backups.
- Network Infrastructure: Secure and high-speed internet connectivity.
- User Devices: Desktops, laptops, tablets, and smartphones.

2. Software Requirements

- Backend Framework: Java Spring Boot for handling business logic and APIs.
- Frontend Framework: JSP for dynamic content rendering.
- Database System: MySQL for structured data management.
- Server: Apache Tomcat for hosting the web application.
- Operating System: Linux-based (Ubuntu) for server efficiency and security.

3. Algorithms Used

- Authentication Algorithm: Secure Hash Algorithm (SHA-256) for password encryption.
- Recommendation Algorithm: Machine learning-based AI for hoarding ad placements.
- Payment Gateway Security: AES encryption to secure financial transactions.

4. Methodology

- Agile Development: Ensures continuous improvement and feedback incorporation.
- RESTful API Integration: Facilitates communication between frontend and backend components.
- Cloud Deployment: Utilizes cloud platforms like AWS or Google Cloud for scalability.

5. Techniques & Approaches

- Data Encryption and Security Measures: Protects user and financial data.
- Load Balancing & Optimization: Ensures high performance and scalability.
- SEO Optimization: Enhances digital reach and advertisement efficiency.
- Automated Testing: Ensures bug-free deployment and high system reliability.

This system, built with Java Spring Boot, JSP, MySQL, and Tomcat, provides a structured and efficient way to manage hoardings, ensuring ease of use and streamlined booking and payment processes.

CHAPTER:5 SYSTEM DESIGN

The system is designed to manage hoardings, users, bookings, and payments efficiently. It includes several key modules:

5.1 System Design & Methodology

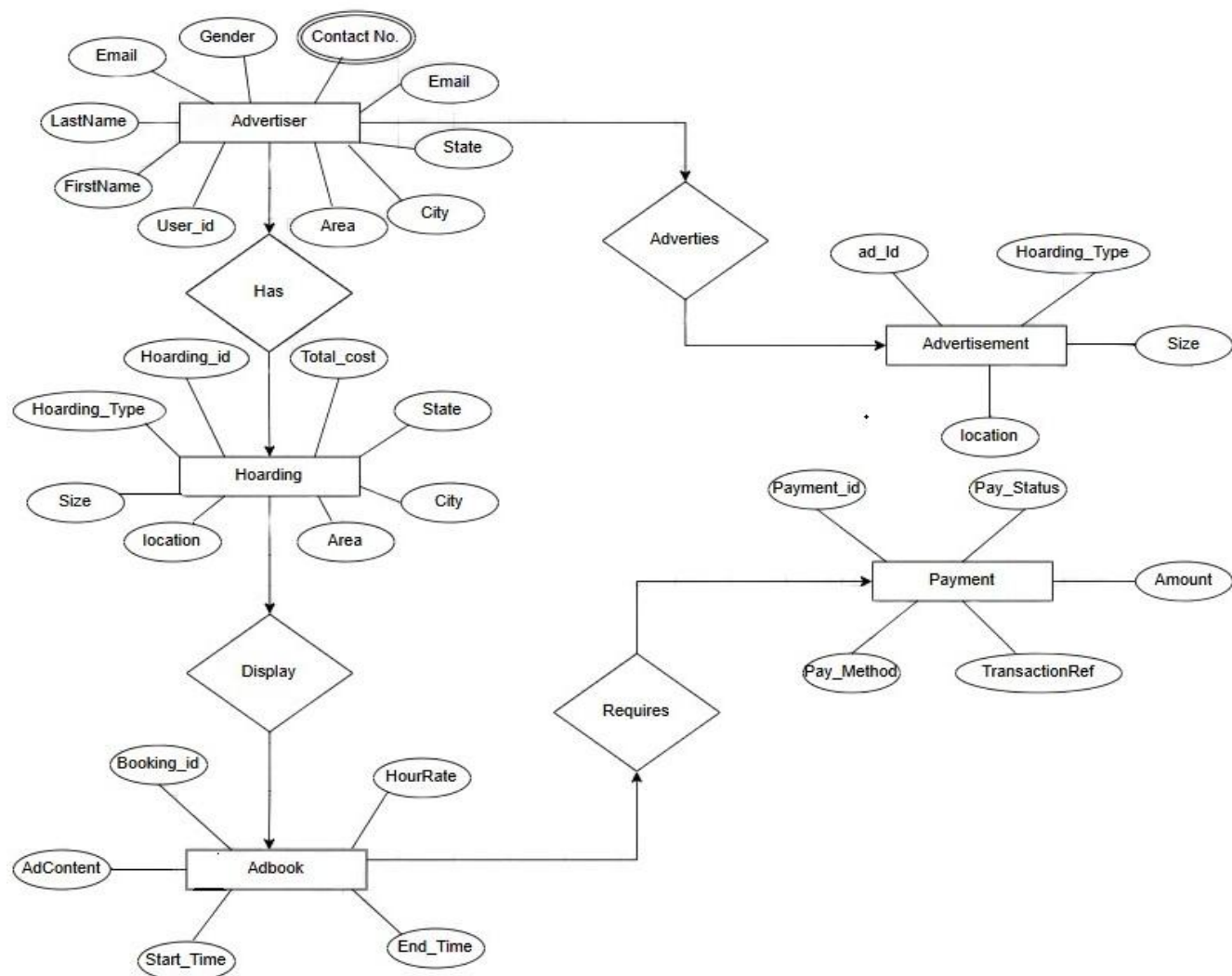


Fig 5.1.1 ER Diagram

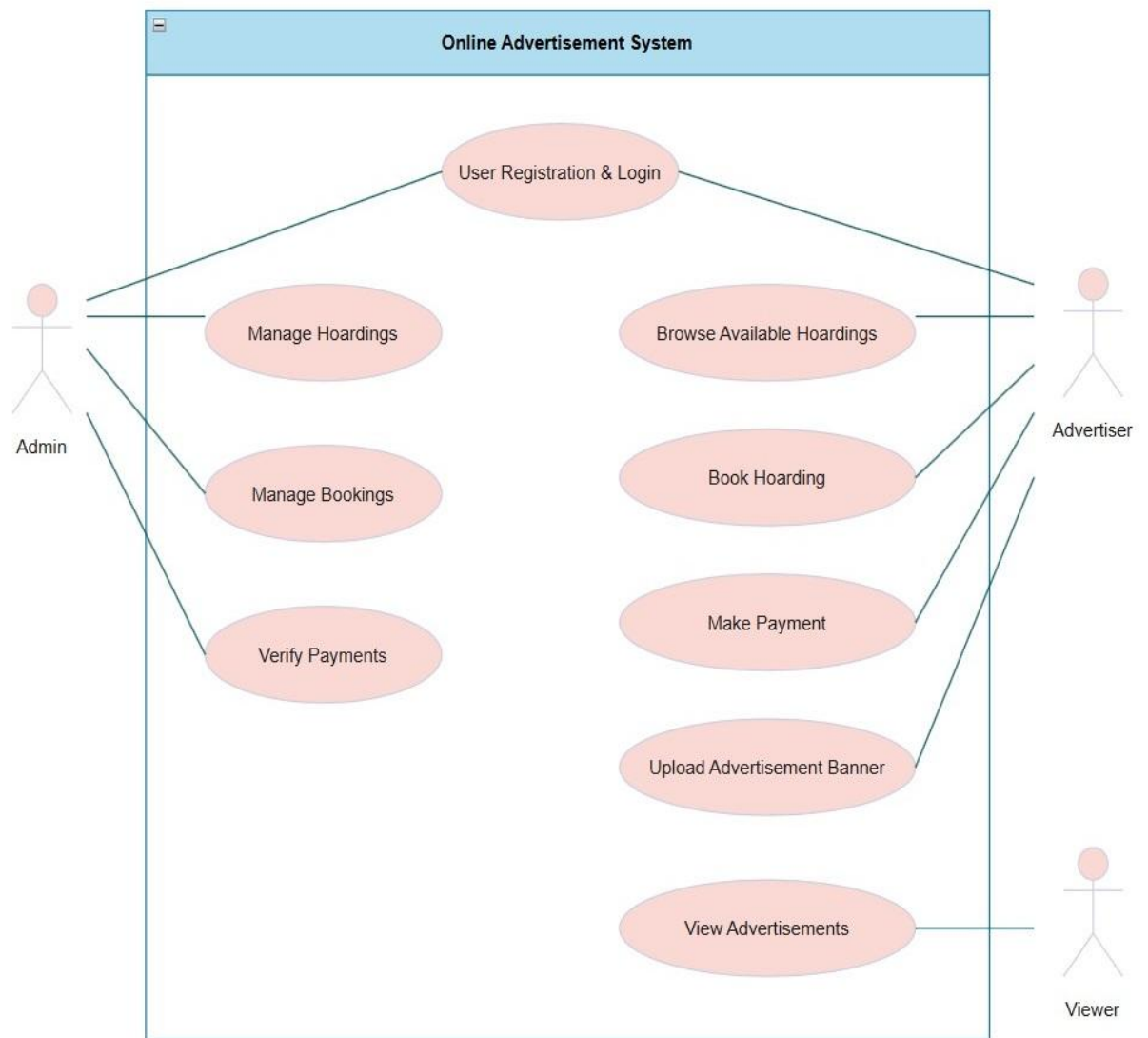


Fig 5.1.2 Use Case Diagram

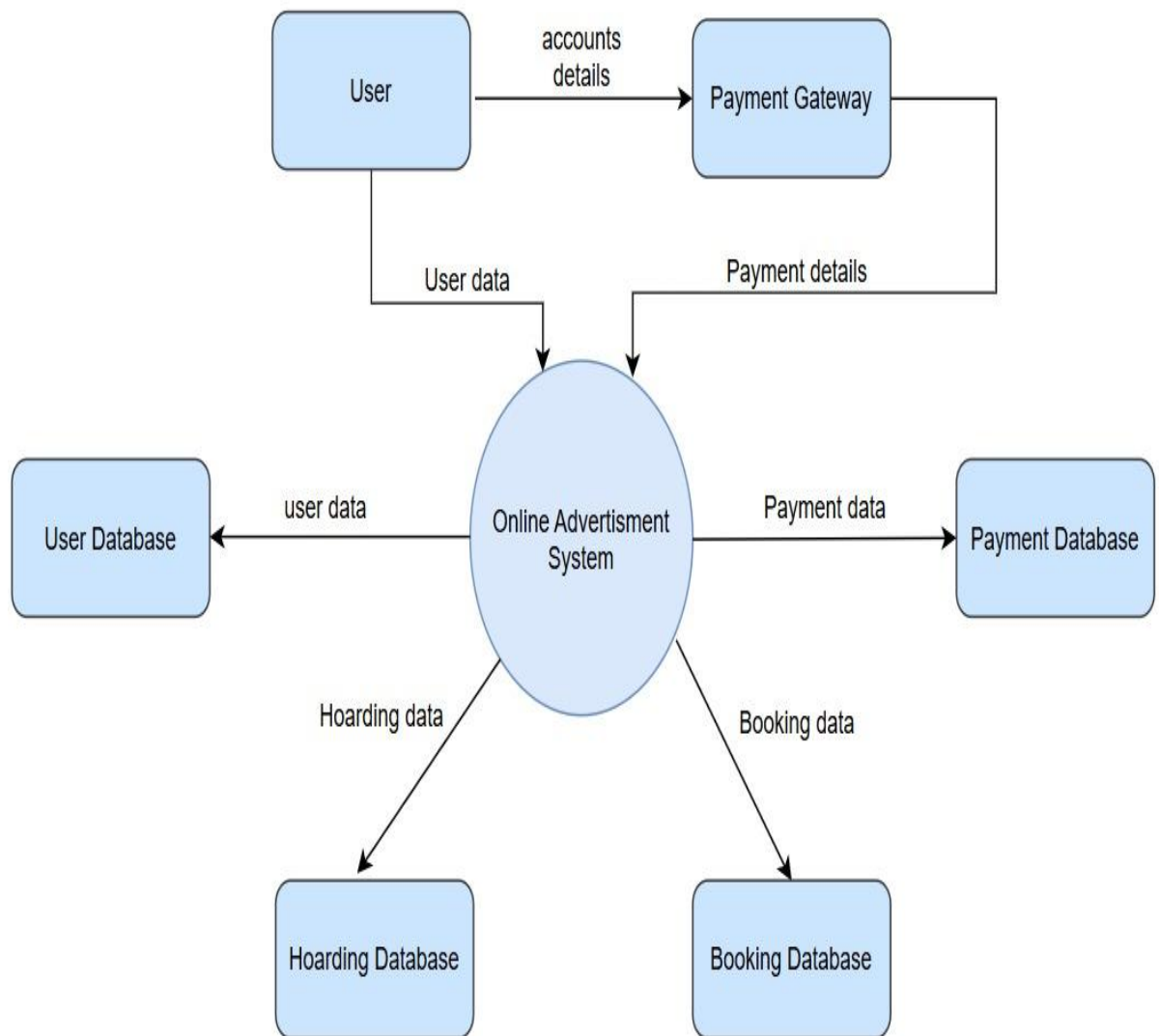


Fig 5.1.3 DFD Level-0 Diagram

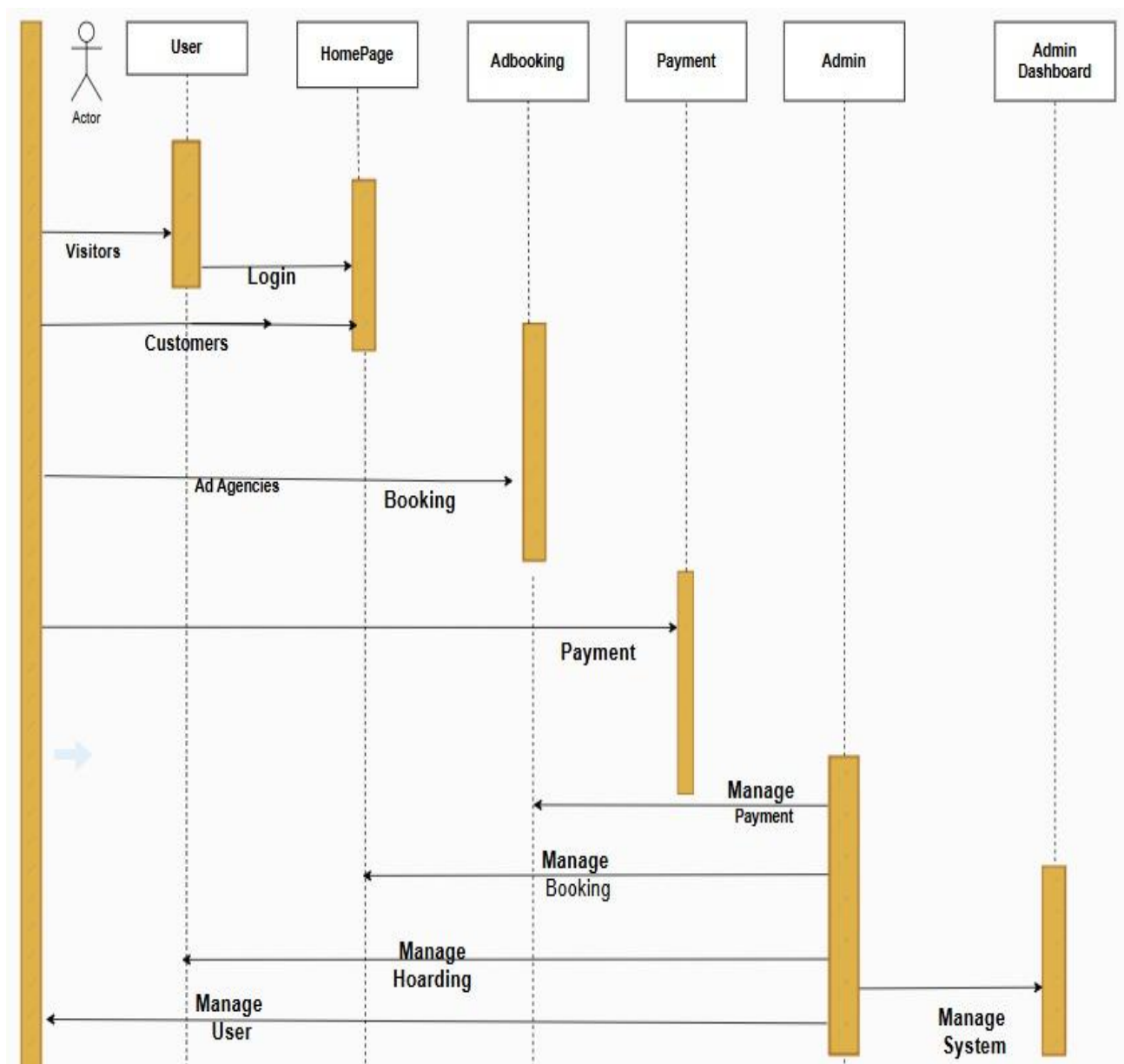
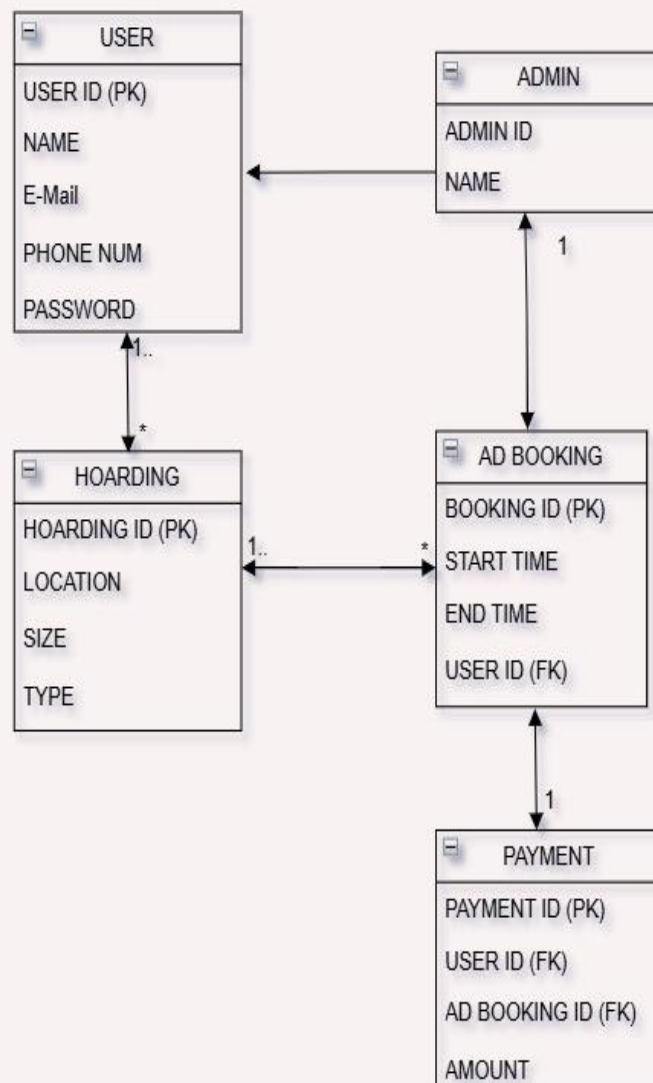
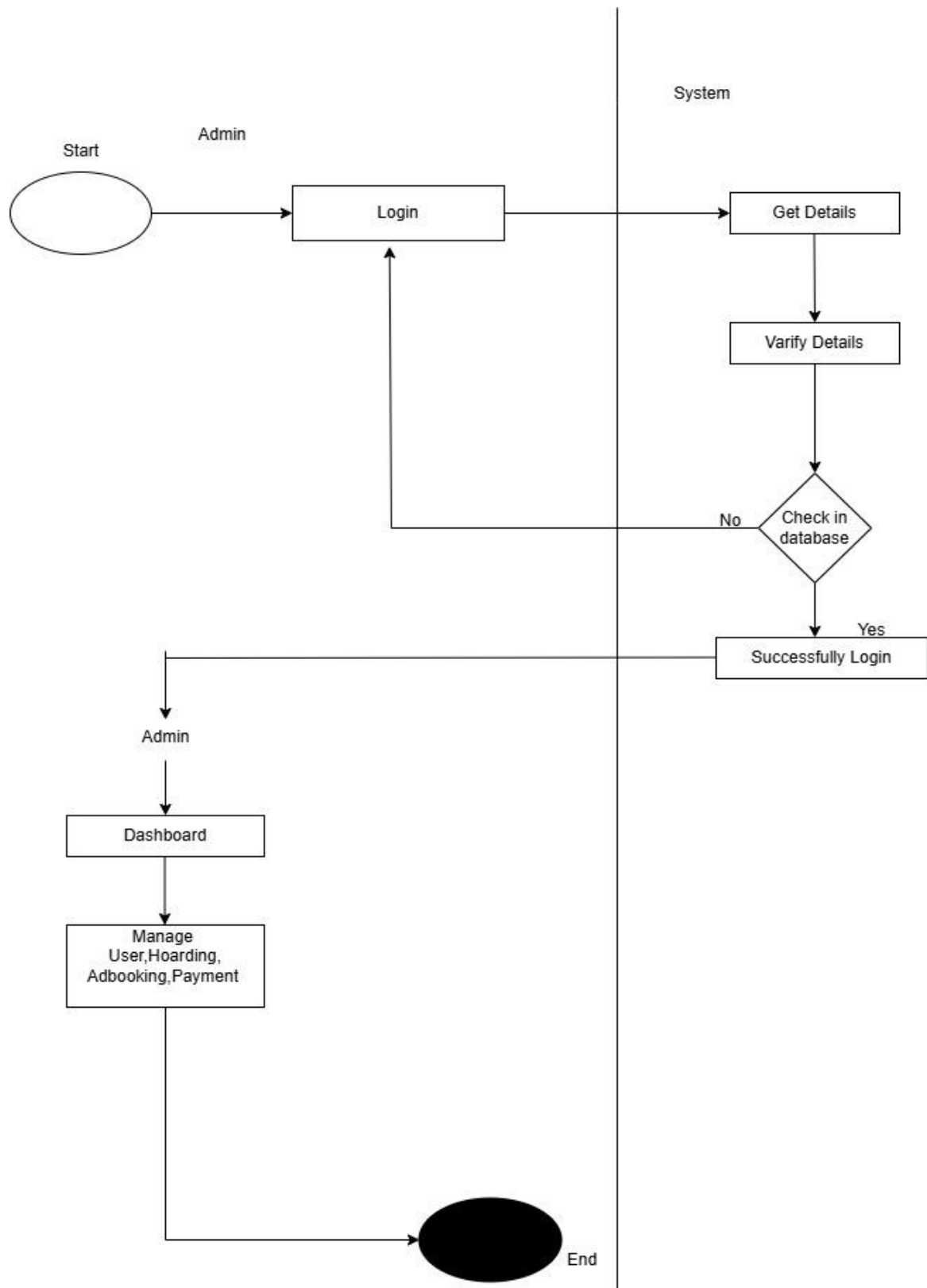


Fig 5.1.4 Sequence Diagram

**Fig 5.1.5 UML Diagram**

**Fig 5.1.6 Activity Diagram**

❖ Database Design

Table 5.1.4 User Table

Table : users		
Field	Type	Constraints
user_id	int	NOT NULL (Primary key) , Auto-increment
password	varchar(255)	NULL
contact_num	varchar(255)	NULL
email	varchar(255)	NULL (UNIQUE)
first_name	varchar(255)	NULL
last_name	varchar(255)	NULL
role	varchar(255)	NULL
gender	varchar(255)	NULL
otp	varchar(255)	NULL
profile_pic_path	varchar(255)	NULL
active	varchar(255)	NULL
create_at	varchar(255)	NULL

Table 5.1.5 Hoarding Table

Table : hoarding		
Field	Type	Constraints
hoardingid	int	NOT NULL (Primary key) , Auto-increment
area	varchar(255)	NULL
banner_url	varchar(255)	NULL
city	varchar(255)	NULL
hoardingtype	varchar(255)	NULL
location	varchar(255)	NULL
pricepertime	varchar(255)	NULL
size	varchar(255)	NULL
state	varchar(255)	NULL
status	varchar(255)	NULL
template_img_path	varchar(255)	NULL

Table 5.1.6 Payment Table

Table : payment		
Field	Type	Constraints
paymentid	int	NOT NULL (Primary key) , Auto-increment
amount	varchar(255)	NULL
paymentmethod	varchar(255)	NULL
transaction_ref	varchar(255)	NULL
user_id	int	NULL
ad_id	int	NULL
paymentstatus	varchar(255)	NULL

Table 5.1.7 State Table

Table : state		
Field	Type	Constraints
state_name	varchar(255)	NULL
stateid	int	NOT NULL (Primary key) , Auto-increment

Table 5.1.8 City Table

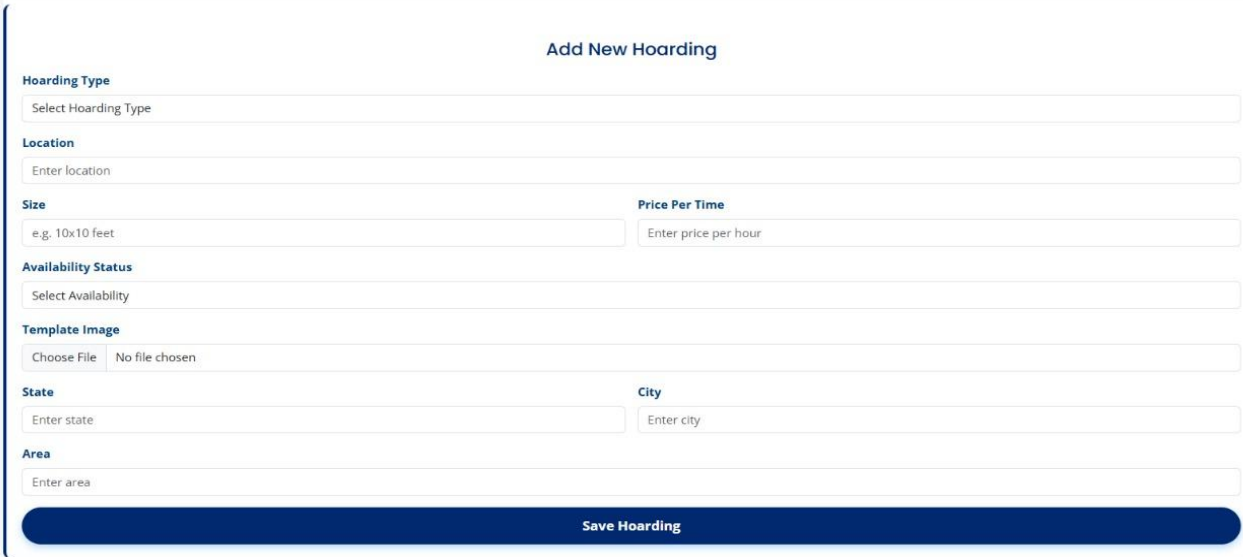
Table : city		
Field	Type	Constraints
cityname	varchar(255)	NULL
cityid	int	NOT NULL (Primary key) , Auto-increment
stateid	int	NULL

Table 5.1.9 Area Table

Table : area		
Field	Type	Constraints
areaname	varchar(255)	NULL
areaid	int	NOT NULL (Primary key) , Auto-increment
cityid	int	NULL
stateid	int	NULL

5.2 Input / Output and Interface Design

5.2.1 Samples of Forms, Reports and Interface



Add New Hoarding

Hoarding Type
Select Hoarding Type

Location
Enter location

Size
e.g. 10x10 feet

Price Per Time
Enter price per hour

Availability Status
Select Availability

Template Image
Choose File No file chosen

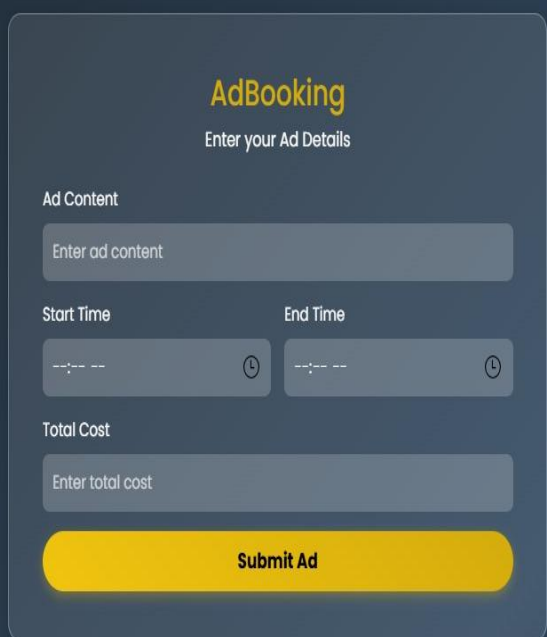
State
Enter state

City
Enter city

Area
Enter area

Save Hoarding

Fig 1 Add Hoarding Form



AdBooking
Enter your Ad Details

Ad Content
Enter ad content

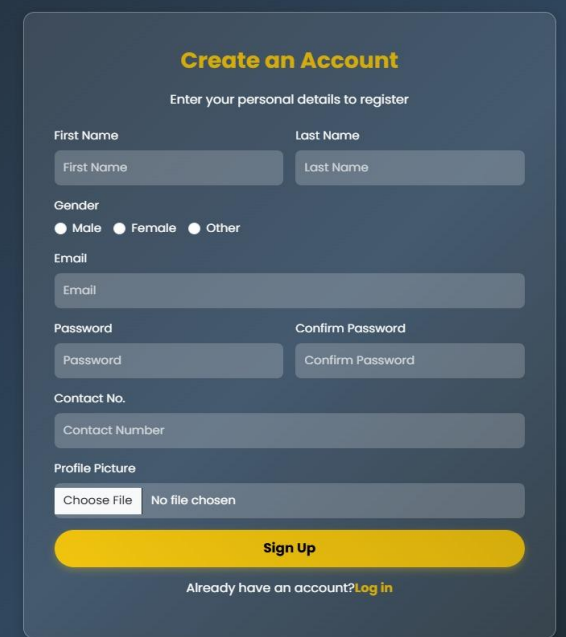
Start Time **End Time**

---:--:-- ⌚ ---:--:-- ⌚

Total Cost
Enter total cost

Submit Ad

Fig 2 Ad Booking Form



Create an Account
Enter your personal details to register

First Name **Last Name**

First Name Last Name

Gender
☐ Male ☐ Female ☐ Other

Email
Email

Password **Confirm Password**

Password Confirm Password

Contact No.
Contact Number

Profile Picture
Choose File No file chosen

Sign Up

Already have an account? [Log in](#)

Fig 3 Signup Form

My Bookings			
Ad Content	Start Time	End Time	Total Price
Online ad	01:06	04:08	3200
Online adver	23:41	14:41	3200
Gucci purse	20:05	21:05	20000
nike shoes	20:30	21:30	10000
chanel foundation	20:50	18:50	3200
chanel lipstick	18:55	21:54	5000
nike shoes	11:44	11:44	10000
nike shoes	16:16	19:17	10000
chanel campact	17:21	19:21	558568
chanel campact	16:24	19:24	558568

Fig 4 My Booking Report

List AdBooking						
Home / List AdBooking						
AdBooking / all						
10	Search:					
entries per page						
Ad content	Start Time	End Time	Total Cost	First Name	Last Name	Actions
adcont	12:32	13:32	20000	shraddha	nigam	  
adcont	14:18	09:33	20000	Madiha	Shaikh	  
adcont	01:47	15:28	20000	Madiha	Shaikh	  
chanel campact	17:21	19:21	558568	diksha	patel	  
chanel campact	16:24	19:24	558568	diksha	patel	  
chanel campact	12:06	14:06	558568	ayana	sharma	  
chanel foundation	20:50	18:50	3200	diksha	patel	  
chanel lipstick	18:55	21:54	5000	diksha	patel	  
Gucci purse	20:05	21:05	20000	diksha	patel	  
igate	12:42	12:46	3200	Madiha	Shaikh	  
Showing 1 to 10 of 20 entries						  1  

Fig 5 AdBooking

CHAPTER:6 IMPLEMENTATION

6.1 Implementation Platform / Environment

- Backend: Java Spring Boot
- Frontend: JSP
- Database: MySQL
- Server: Tomcat

6.2 Process / Program / Technology / Modules Specification(s)

- Booking Module: Manages reservations and scheduling.
- Payment Module: Handles transactions securely.
- Advertisement Module: Manages ad uploads and approvals.
- User Management Module: Controls authentication and roles.
- Notification Module: Sends alerts and updates via email and SMS.
- Digital Hoarding Management: Supports remote and automated content updates.

6.3 Finding / Results / Outcomes

Welcome to SpotAdz
Advertise Smart, Reach Far! Join our platform and showcase your ads like never before.
"SpotAdz: Join Today & Expand Your Advertising Potential! Your Growth Starts Here."

[Login](#)

Create an Account

Enter your personal details to register

First Name Last Name

Gender
☐ Male ☐ Female ☐ Other

Email

Password Confirm Password

Contact No.

Profile Picture
 No file chosen

[Sign Up](#)

Already have an account? [Log In](#)

Welcome to SpotAdz
Advertise Smart, Reach Far! Join our platform and showcase your ads like never before.
"SpotAdz: Join Today & Expand Your Advertising Potential! Your Growth Starts Here."

[Signup](#)

Login Here

Email

Password

[Log In](#)

[Forgot password?](#)

Create an account? [Signup](#)

Forget Password
Enter your email to change password

Email

[Reset Password](#)

[Back to Log In](#)

Change Password
Set a new password

Email

OTP

New Password

Confirm Password

[Change Password](#)

[Back to Log In](#)

HOMEPAGE

AD SpotAdz

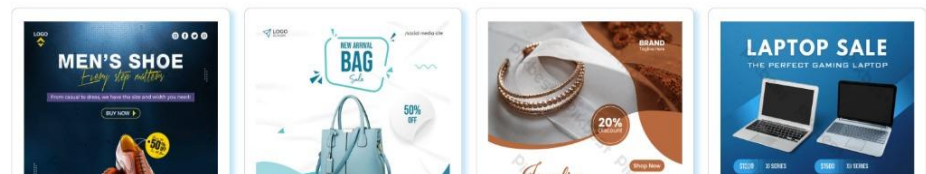
Home Ads MyBooking MyPayment About Contact diksha

Advertise Smart, Grow Fast

Join our platform to promote your business and reach the right audience.

[Get Started](#)

Featured Advertisements



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Cosmetic
Premium Natural
50% OFF
Mail
palladium Mall
Size : 20x30
[View Ad](#)

WOMAN FRAGRANCE SALE
up to 50% OFF
Billboard
Nayab Shop
Size : 20x30
[View Ad](#)

SPECIAL WATCH
EXCLUSIVE COLLECTION
UP TO 35% OFF
Billboard
Front of Amul Parlour
Size : 30x50
[View Ad](#)

SMART LED TV
NEW ARRIVAL
Billboard
Bapunagar
Size : 30x50
[View Ad](#)

Engaging Video Templates for Powerful Advertising

Create compelling video ads in minutes! Capture attention, increase conversions, and grow your brand.

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Stylish Men's Shoes - Up to 50% Off
Get the perfect shoes for any occasion. Comfortable, stylish, and affordable!
[View Ad](#)

Luxury Perfume Sale - 50% Off
Find your perfect scent! Long-lasting and refreshing perfumes at great prices.
[View Ad](#)

Special Beauty
envato
Special Beauty

FOOD NAME

ADMIN DASHBOARD

SpotAdz Admin Dashboard

Dashboard (Home / Dashboard)

- Users (Overall)**: 17
- Hoardings (Total)**: 8
- Monthly Users (APRIL)**: 2

About Us
At SpotAdz, we help businesses grow with smart and effective online advertising. Our platform makes it easy to reach the right audience and boost visibility.

Contact Us
+91 9756787126
+91 9828593083
+81 83209 3502

Follow Us
f t i g

SpotAdz Admin Dashboard

Shaikh Noman

First Name	Shaikh
Last Name	Noman
Gender	male
Email	shaikhnoman0652@gmail.com
Contact Number	7567387126
Role	ADMIN
Account Status	Active
Joined On	2025-04-05

SpotAdz Admin Dashboard

Add New Hoarding

Hoarding Type
Select Hoarding Type

Location
Enter location

Size
e.g. 10x10 feet

Price Per Hour
Enter price per hour

Availability Status
Select Availability

Template Image
Choose File | No file chosen

State
Enter state

City
Enter city

Area
Enter area

Save Hoarding

AD SpotAdz

Search

Admin

Users

States

Cities

Area

Hoarding

AdBooking

Payment

List User

Home / List User

Users / all

10

Search:

entries per page

User Id	First Name	Last Name	Email	Gender	Contact Num	Action
1	Shaikh	Avesh	Avesh786@yopmail.com	male	7567387126	
2	Shaikh	Avesh	Avesh786@yopmail.com	male	7567387126	
3	Shaikh	Avesh	Avesh786@yopmail.com	male	7567387126	
4	Shaikh	Avesh	Avesh786@yopmail.com	male	7567387126	
5	Shaikh	Avesh	Avesh786@yopmail.com	male	7567387126	
6	Shaikh	Avesh	Avesh786@yopmail.com	male	7567387126	
7	Shaikh	Noman	shaikhnoman0652@gmail.com	male	7567387126	
8	Shaikh	Atik	atik786@yopmail.com	male	7567387126	
10	Saiyed	Akil	saiyadakil44@gmail.com	male	7567387126	

Showing 1 to 9 of 9 entries

1

2

3

4

5

AD SpotAdz

Search

Admin

Users

States

Cities

Area

Hoarding

AdBooking

Payment

View User

Home / View User

First Name: Shaikh

Last Name: Avesh

Email: Avesh786@yopmail.com

Gender: male

Contact Num: 7567387126

AD SpotAdz

Search

Admin

Users

States

Cities

Area

Hoarding

AdBooking

Payment

Edit User's Details

First Name

Shaikh

Last Name

Avesh

Email

Avesh786@yopmail.com

Contact No.

7567387126

Profile Pic

Choose File No file chosen

Update User

Back to List

6.4 Result Analysis / Comparison / Deliberations

- Performance Benchmarking: Analysis of booking speed, system uptime, and response times.
- Security Assessments: Comparing actual security performance with industry standards.
- Cost Efficiency: Evaluating ROI based on revenue growth and operational cost reduction.
- User Feedback Comparison: Insights from customer reviews and satisfaction ratings.
- Competitor Analysis: Benchmarking against similar platforms to identify unique advantages.

CHAPTER:7 TESTING

7.1 Testing Strategies

Unit Testing: Individual components tested using JUnit.

Integration Testing: Ensures modules interact correctly.

System Testing: Verifies end-to-end functionalities.

Performance Testing: Measures system efficiency under load.

Security Testing: Validates data protection measures.

User Acceptance Testing (UAT): Ensures functionality meets business requirements.

7.2 Test Results and Analysis

7.2.1 Test Cases

Test ID: TC01

Test Condition: User login with valid credentials

Expected Output: Login successful

Actual Output: Login successful

Remark: Pass

Test ID: TC02

Test Condition: Booking a hoarding with correct details

Expected Output: Booking confirmed

Actual Output: Booking confirmed

Remark: Pass

Test ID: TC03

Test Condition: Payment with invalid card details

Expected Output: Payment declined

Actual Output: Payment declined

Remark: Pass

Test ID: TC04

Test Condition: Searching for available hoardings

Expected Output: List of available hoardings displayed

Actual Output: List displayed

Remark: Pass

Test ID: TC05

Test Condition: Unauthorized user trying to access admin panel

Expected Output: Access denied

Actual Output: Access denied

Remark: Pass

CHAPTER:8 CONCLUSION AND DISCUSSION

This section provides a comprehensive overview of the internship/project, including an evaluation of its viability, documented progress through reviews and mentor visits, challenges faced, and potential future improvements.

8.1 Overall Analysis of Internship / Project Viabilities

The internship/project was highly viable and impactful, successfully achieving its primary objectives of managing hoarding bookings and advertisement placements efficiently. The system demonstrated strong capabilities in automation, security, and scalability, making it a practical solution for real-world implementation.

Key Viability Factors:

- **Technical Feasibility:** The project was implemented using robust backend and frontend technologies, ensuring a stable and scalable system.
- **Operational Feasibility:** The system effectively reduced manual booking processes, enhancing efficiency and user convenience.
- **Financial Feasibility:** The solution minimizes operational costs through automation, cloud hosting, and self-service functionalities.
- **Market Feasibility:** There is a growing demand for digital advertisement booking systems, making this a commercially viable product.

Conclusion:

The internship/project demonstrated strong potential for real-world deployment and can be further enhanced with advanced features such as AI-driven ad targeting and blockchain-based payments to improve efficiency and security.

8.2 Dates of Continuous Evaluation (Internal Reviews)

The project was continuously evaluated through internal reviews conducted by the faculty to track progress, assess challenges, and provide guidance.

Internal Review 1: [08/02/2025] Evaluated project scope, initial system design, and feasibility. Feedback was given on improving backend and data

Internal Review 2: [05/04/2025] Assessed final-stage development, module integrations, and security aspects. Recommendations were provided for UI Design.

8.3 Problems Encountered and Possible Solutions

During the development and implementation of the project, several challenges were encountered. Below is a summary of key problems and the solutions applied:

- **Problem Encountered :** UI/UX complexity for first-time users
- **Solution Implemented :** Redesigned the dashboard with simplified navigation and provided in-app tooltips for guidance.
- **Problem Encountered :** Limited payment options
- **Solution Implemented :** Planned integration of Google Pay and Apple Pay in future updates.

These solutions significantly improved the system's functionality, ensuring a smooth and secure user experience.

8.4 Summary of Internship / Project Work

The internship/project provided valuable hands-on experience in software development, testing, security, and deployment. The system was successfully built and demonstrated efficiency in automating hoarding bookings and advertisement placements.

Key Takeaways from the Project:

Technical Learning:

- Gained expertise in backend development (Java Spring) and frontend frameworks (JSP).
- Learned database management, API integrations, and security best practices.

Project Management Skills:

- Understood the importance of agile development methodologies.
- Learned to handle challenges, debug issues, and optimize performance.

Final Outcome:

- ❖ The project achieved its intended objectives and provided a scalable, secure, and efficient platform for digital advertisement management.

8.5 Limitations and Future Enhancements

Limitations of the Current System

Despite the success of the project, some limitations were identified, which can be addressed in future updates:

Limitation:

- ❖ Limited AI-driven insights (Manual ad targeting may not be as effective).
- ❖ Scalability under extreme load (System may slow down with thousands of concurrent users).
- ❖ Limited Payment Option (Some users may not find their preferred payment method).

Future Enhancements

To further improve the system, the following enhancements are planned:

1. Mobile App Development

- Launch a dedicated mobile application for enhanced user accessibility.

2. Live Digital Hoarding Management

- Enable real-time streaming and remote updates for digital advertisements.

These enhancements will increase the system's efficiency, security, and market competitiveness, making it a leading digital hoarding management platform.

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- <https://chat.openai.com/>
- <https://www.deepseek.com/>